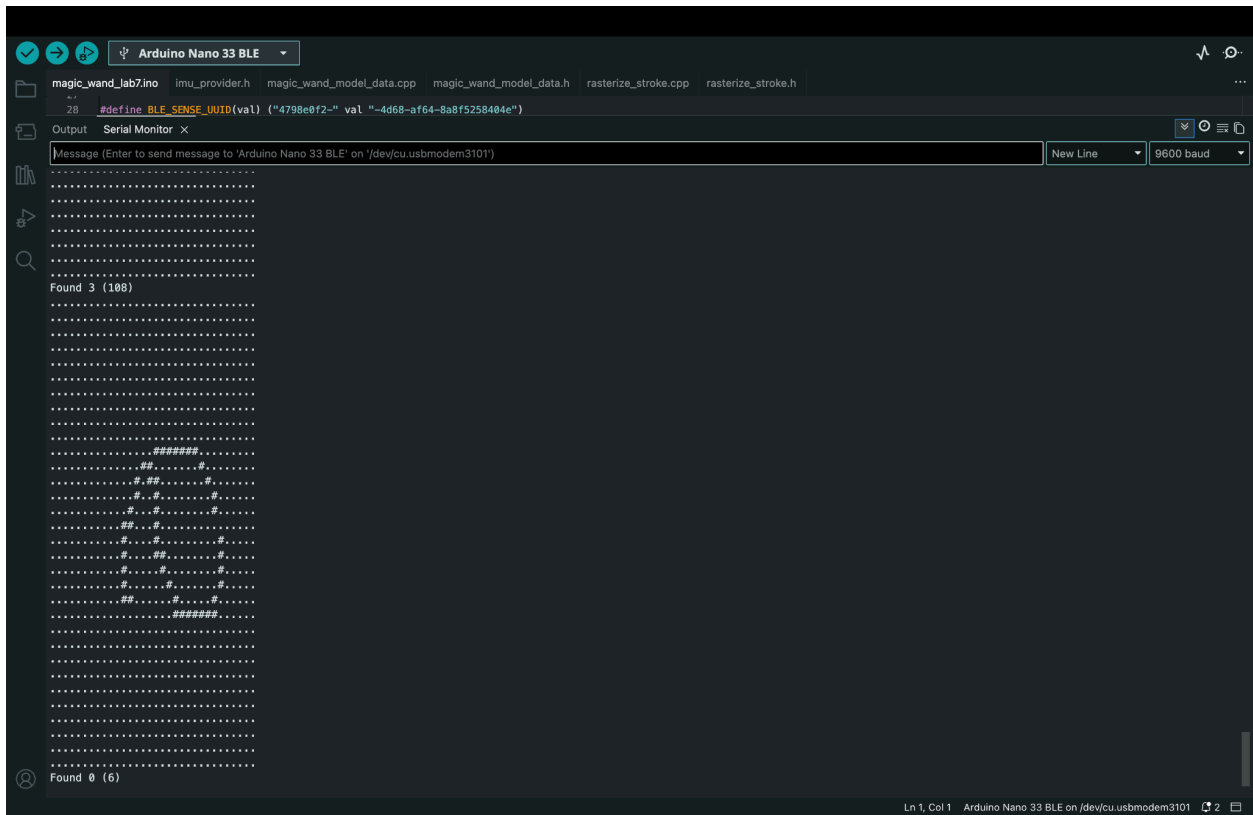


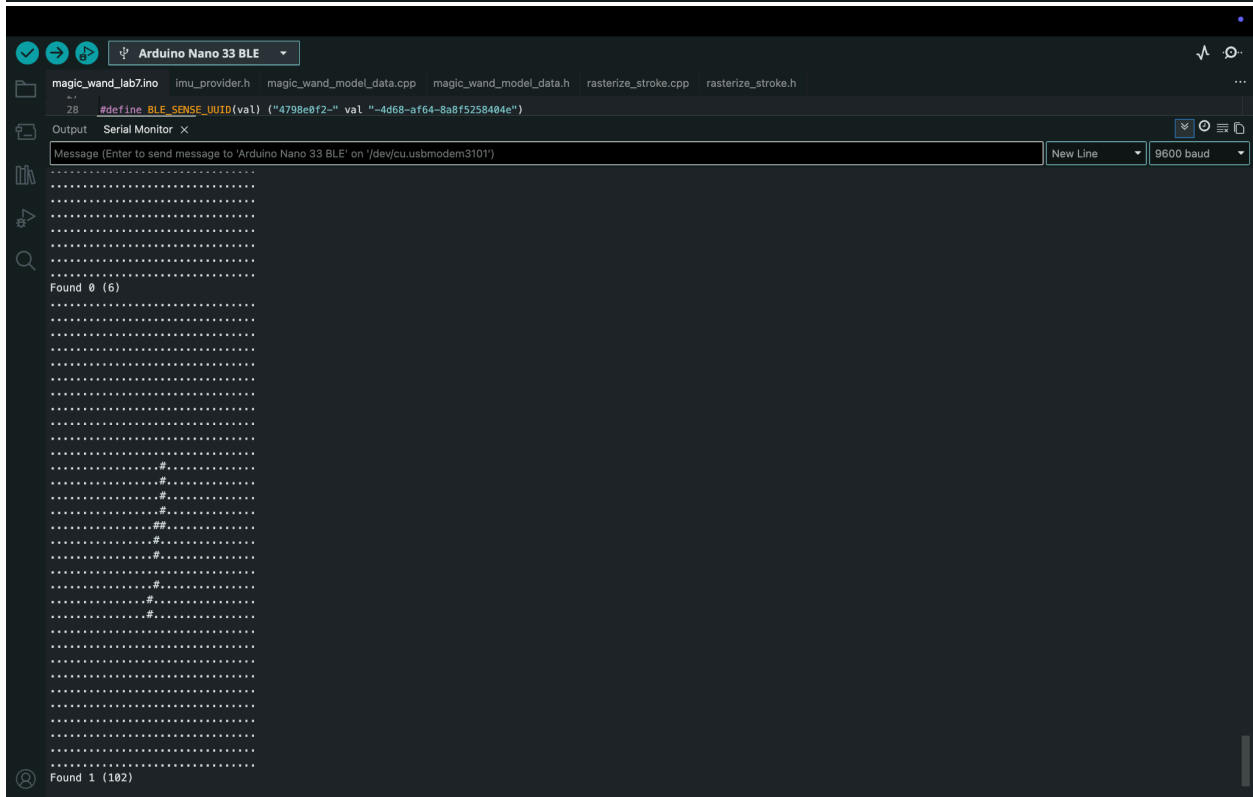
Baseline Model



The screenshot shows the Arduino IDE interface with the 'Arduino Nano 33 BLE' selected. The code editor displays the file 'magic_wand_lab7.ino' with the following content:

```
magic_wand_lab7.ino imu_provider.h magic_wand_model_data.cpp magic_wand_model_data.h rasterize_stroke.cpp rasterize_stroke.h
28 #define BLE_SENSE_UUID(val) ("4798e0f2-" val "-4d68-af64-8a8f5258484e")
```

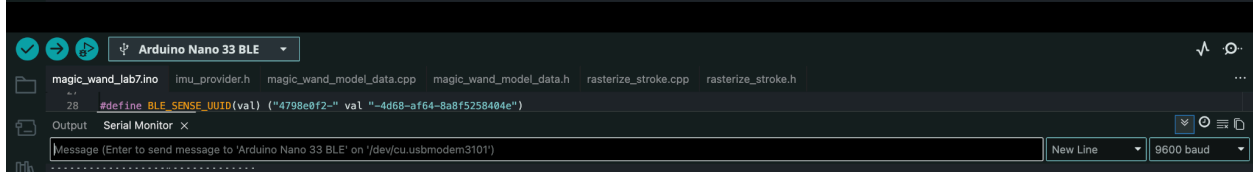
The Serial Monitor is open, showing a message input field and a dropdown menu set to 'New Line' and '9600 baud'. The output area displays a large number of dots, indicating received data. The status bar at the bottom indicates 'Ln 1, Col 1 Arduino Nano 33 BLE on /dev/cu.usbmodem3101'.



The screenshot shows the Arduino IDE interface with the 'Arduino Nano 33 BLE' selected. The code editor displays the file 'magic_wand_lab7.ino' with the following content:

```
magic_wand_lab7.ino imu_provider.h magic_wand_model_data.cpp magic_wand_model_data.h rasterize_stroke.cpp rasterize_stroke.h
28 #define BLE_SENSE_UUID(val) ("4798e0f2-" val "-4d68-af64-8a8f5258484e")
```

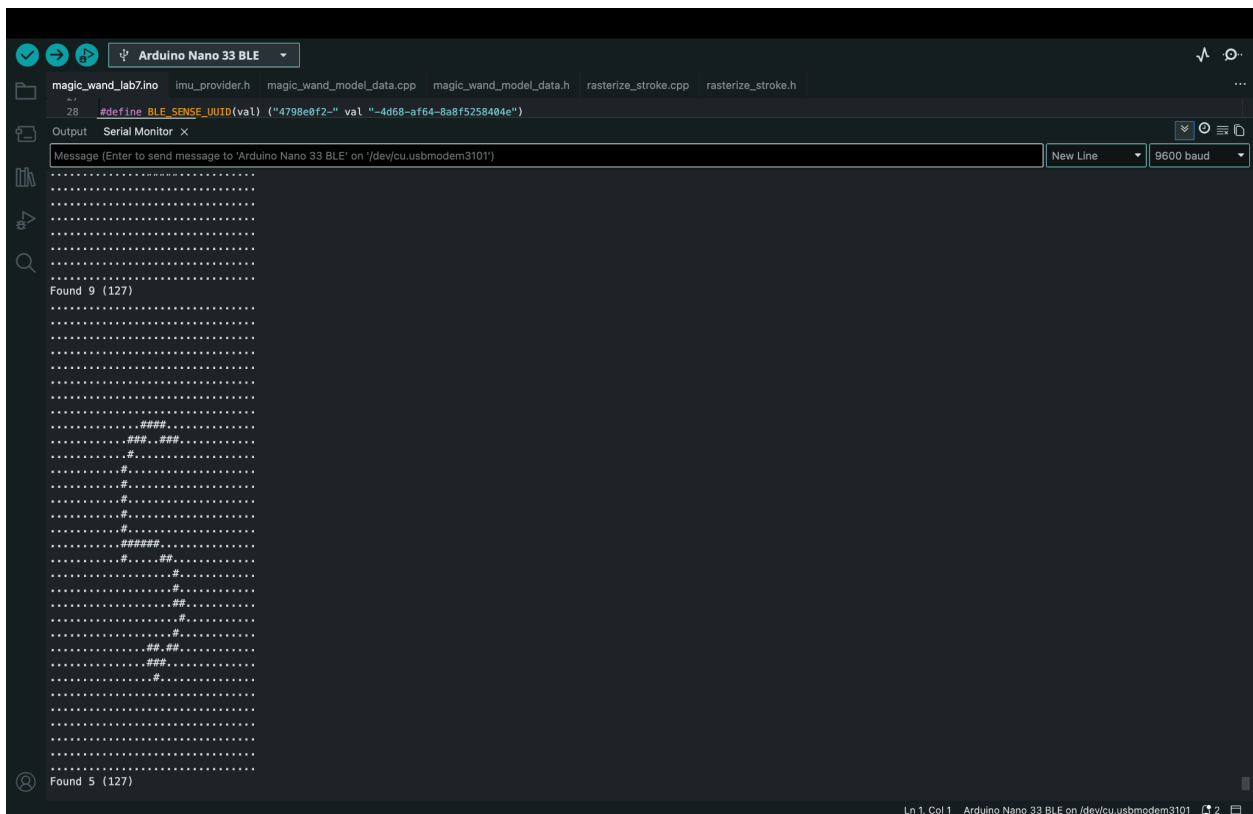
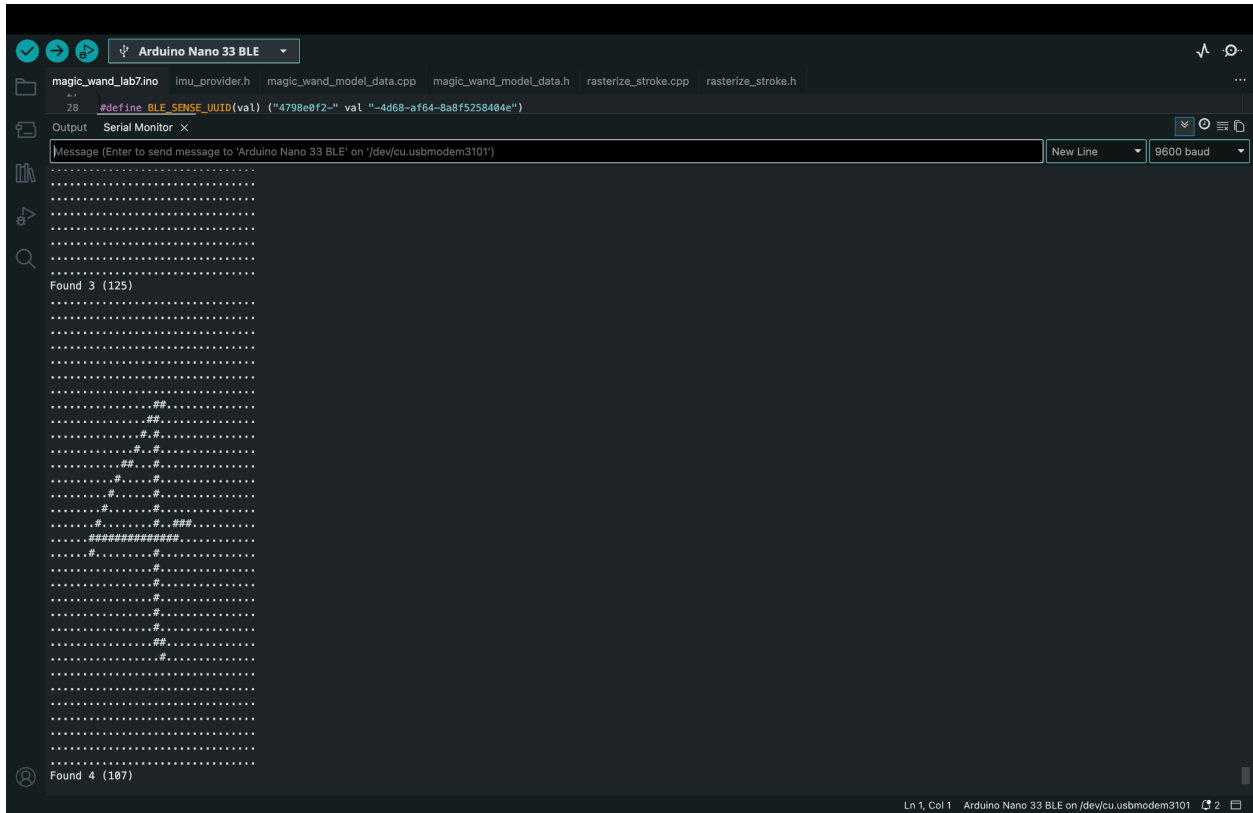
The Serial Monitor is open, showing a message input field and a dropdown menu set to 'New Line' and '9600 baud'. The output area displays a large number of dots, indicating received data. The status bar at the bottom indicates 'Ln 1, Col 1 Arduino Nano 33 BLE on /dev/cu.usbmodem3101'.

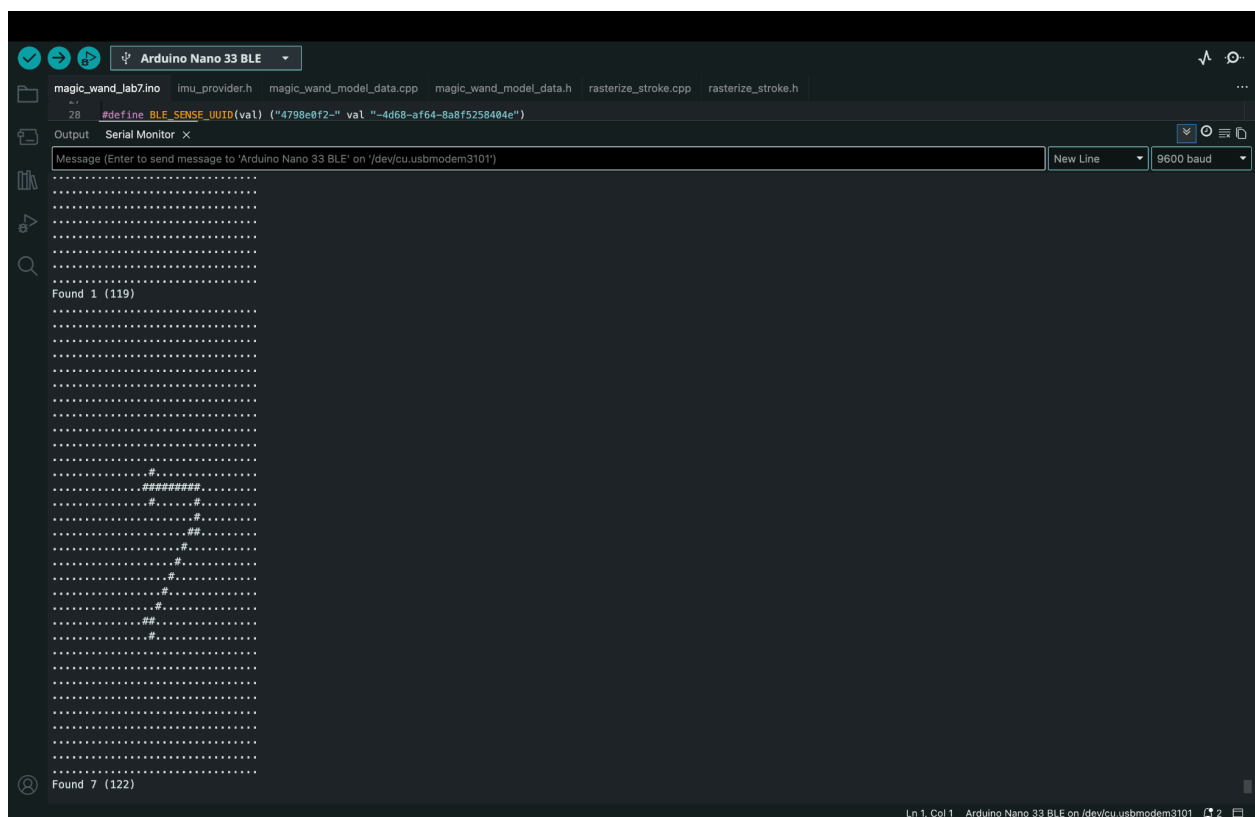
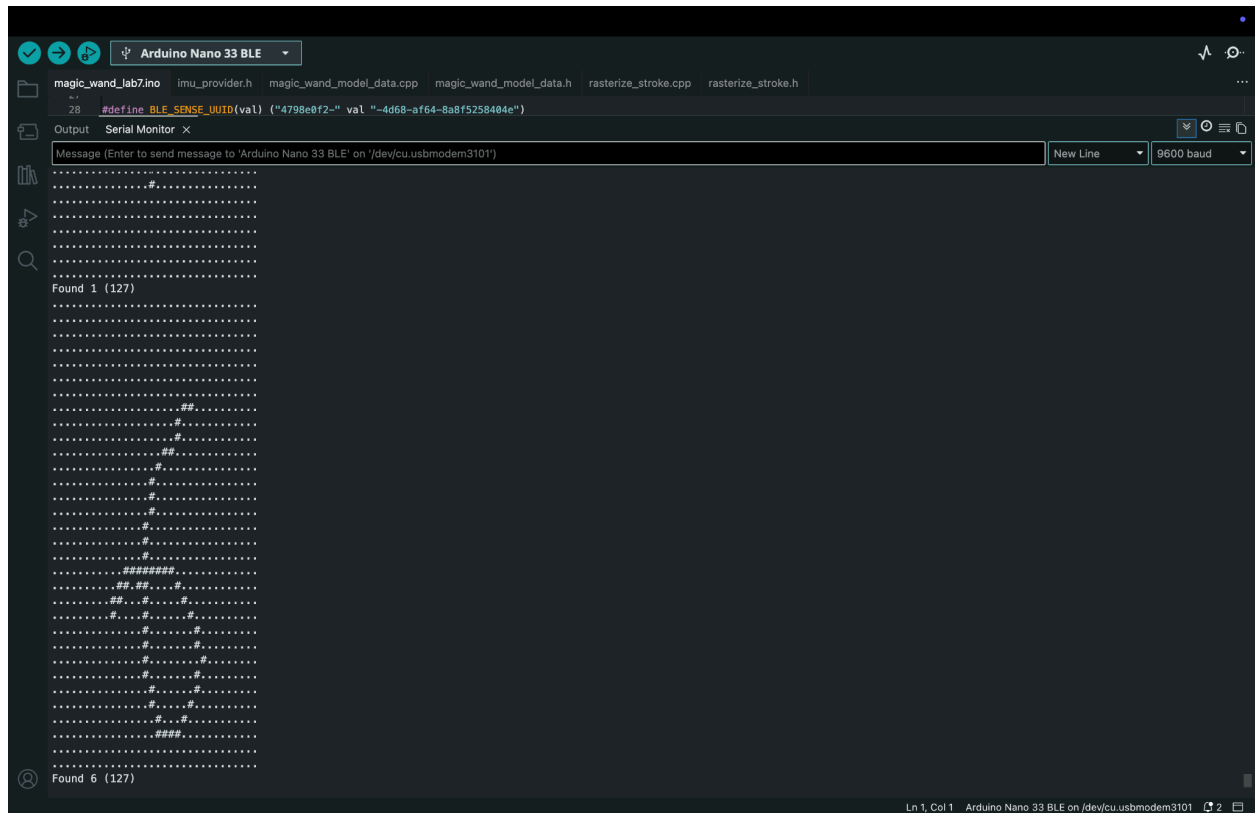


The screenshot shows the Arduino IDE interface with the 'Arduino Nano 33 BLE' selected. The code editor displays the file 'magic_wand_lab7.ino' with the following content:

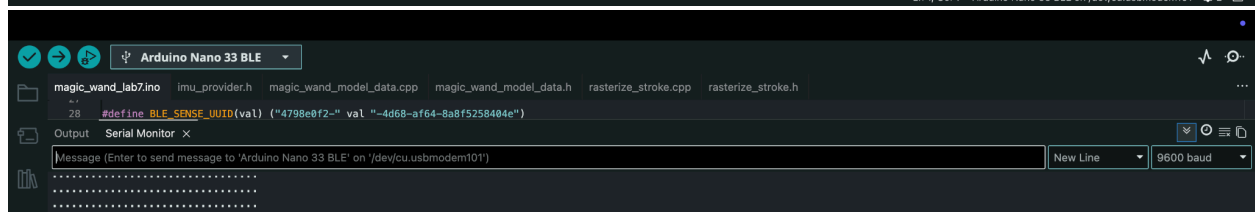
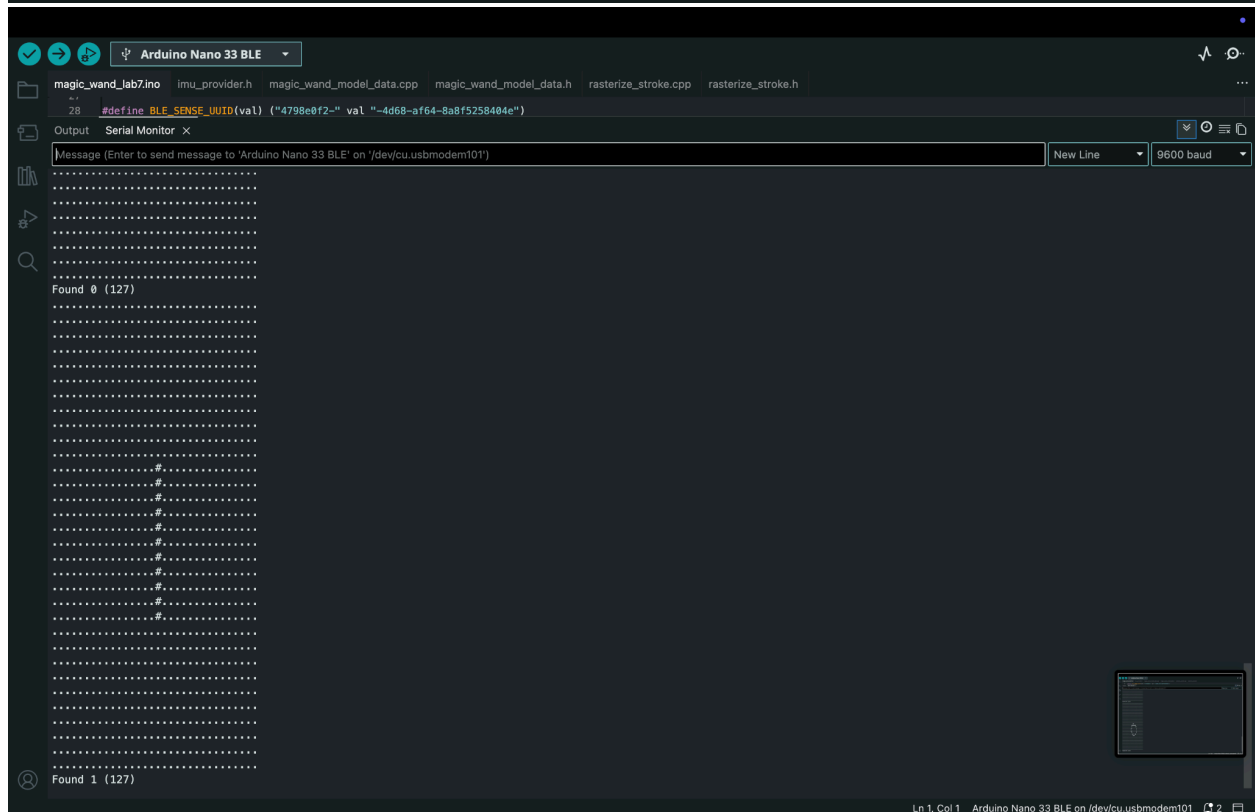
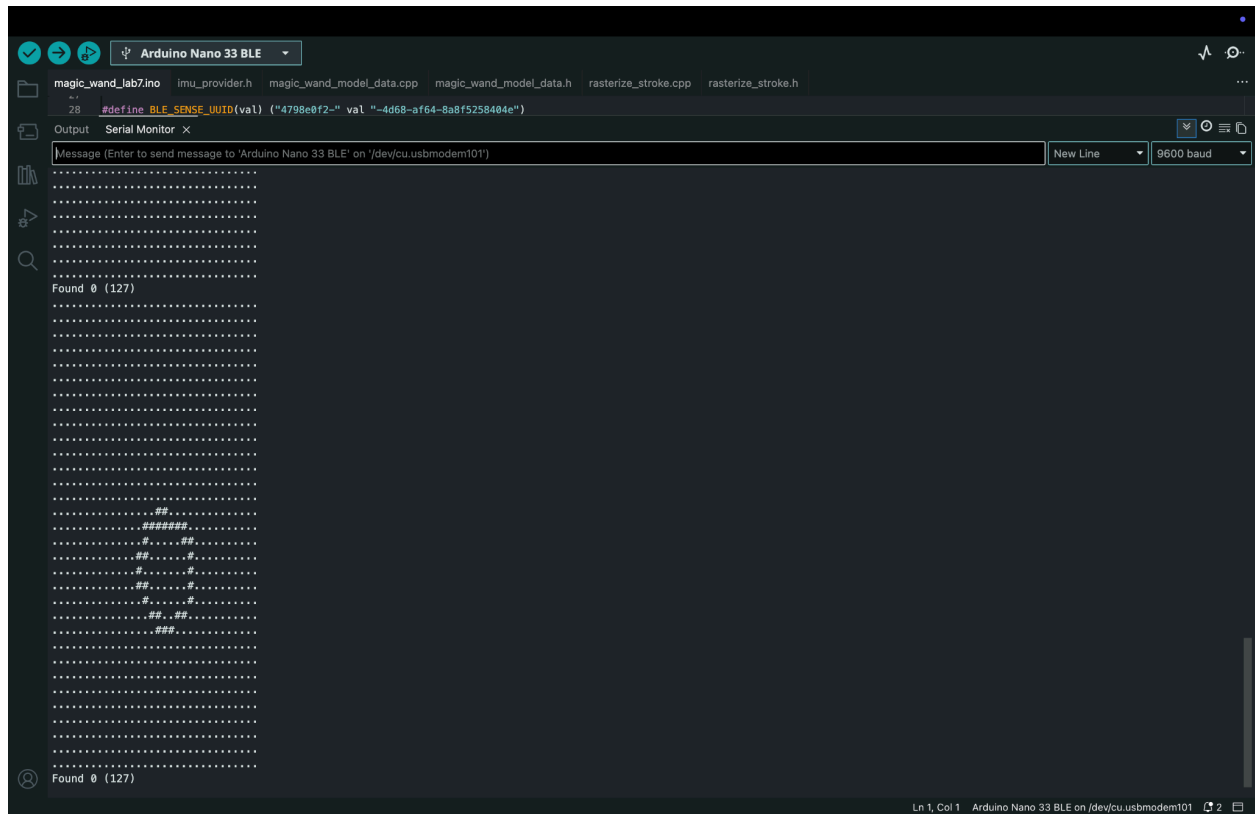
```
magic_wand_lab7.ino imu_provider.h magic_wand_model_data.cpp magic_wand_model_data.h rasterize_stroke.cpp rasterize_stroke.h
28 #define BLE_SENSE_UUID(val) ("4798e0f2-" val "-4d68-af64-8a8f5258484e")
```

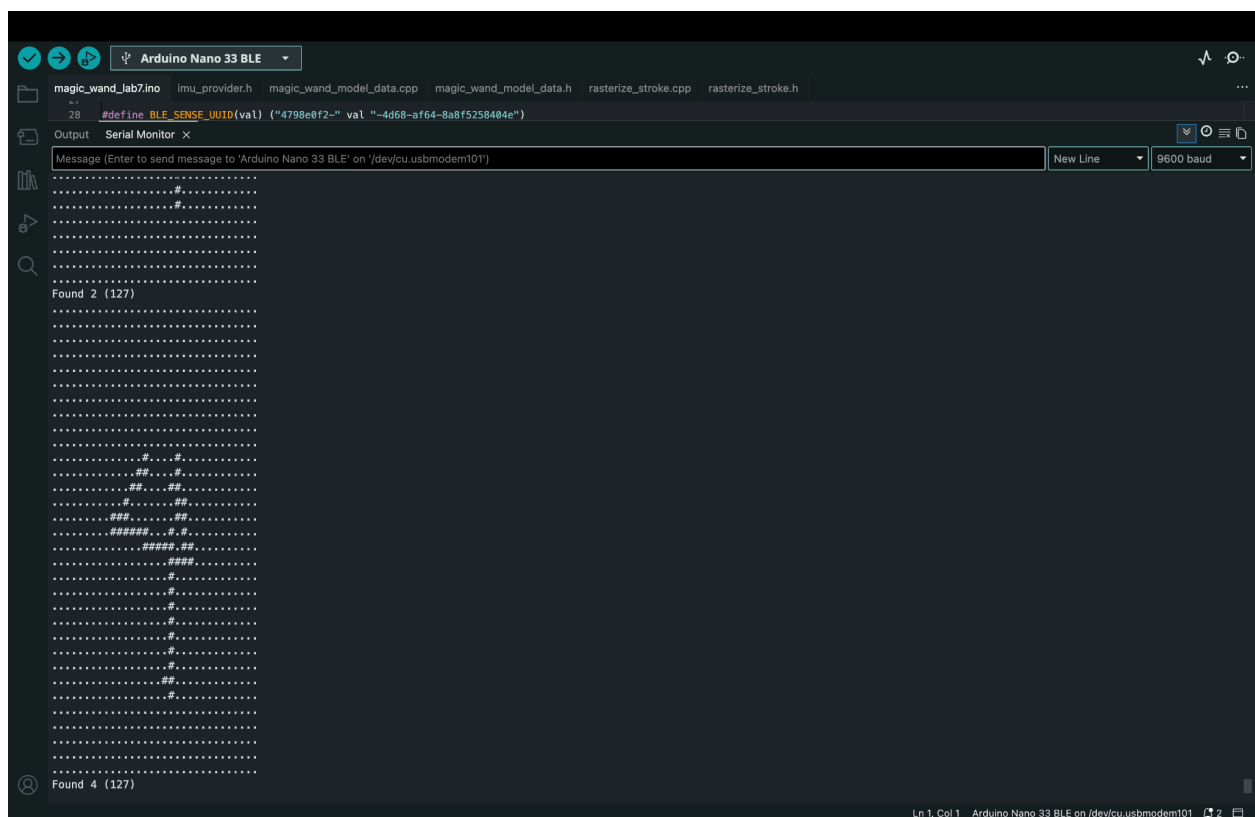
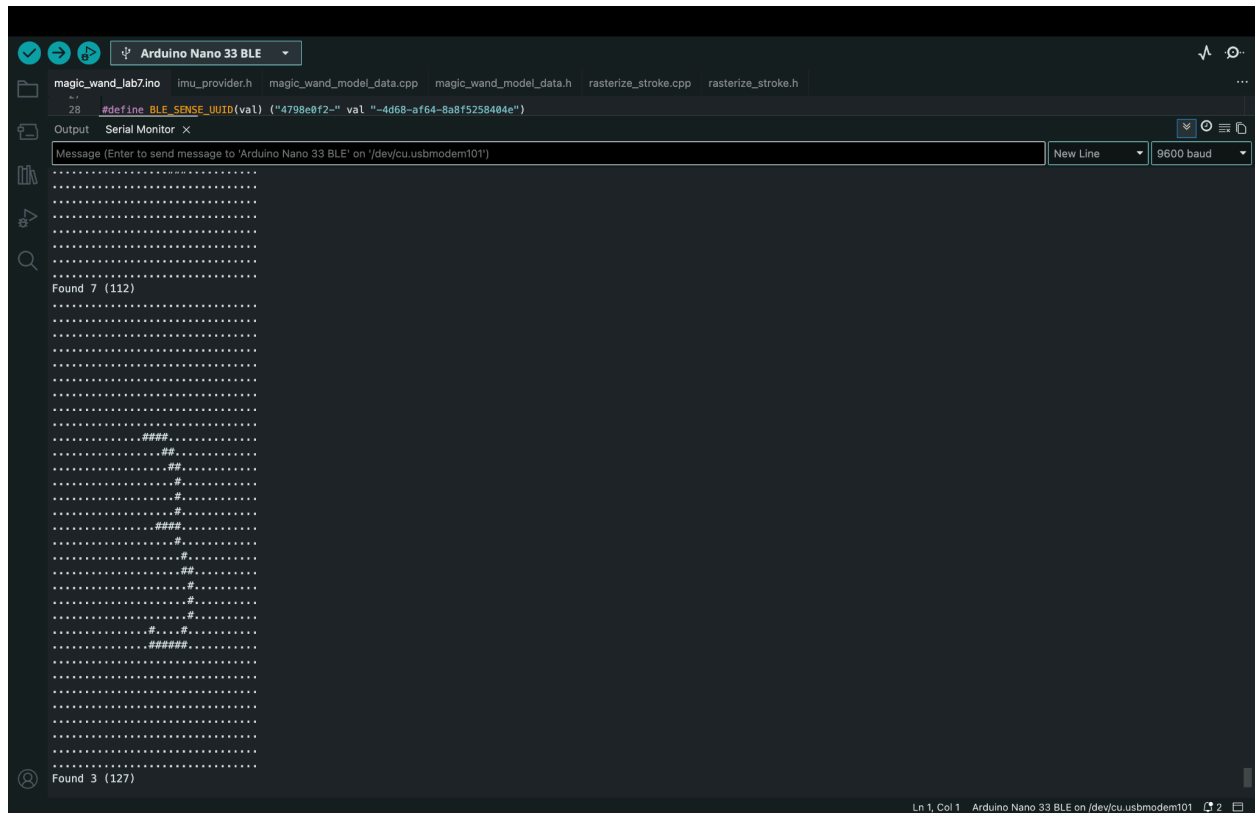
The Serial Monitor is open, showing a message input field and a dropdown menu set to 'New Line' and '9600 baud'. The output area displays a large number of dots, indicating received data. The status bar at the bottom indicates 'Ln 1, Col 1 Arduino Nano 33 BLE on /dev/cu.usbmodem3101'.

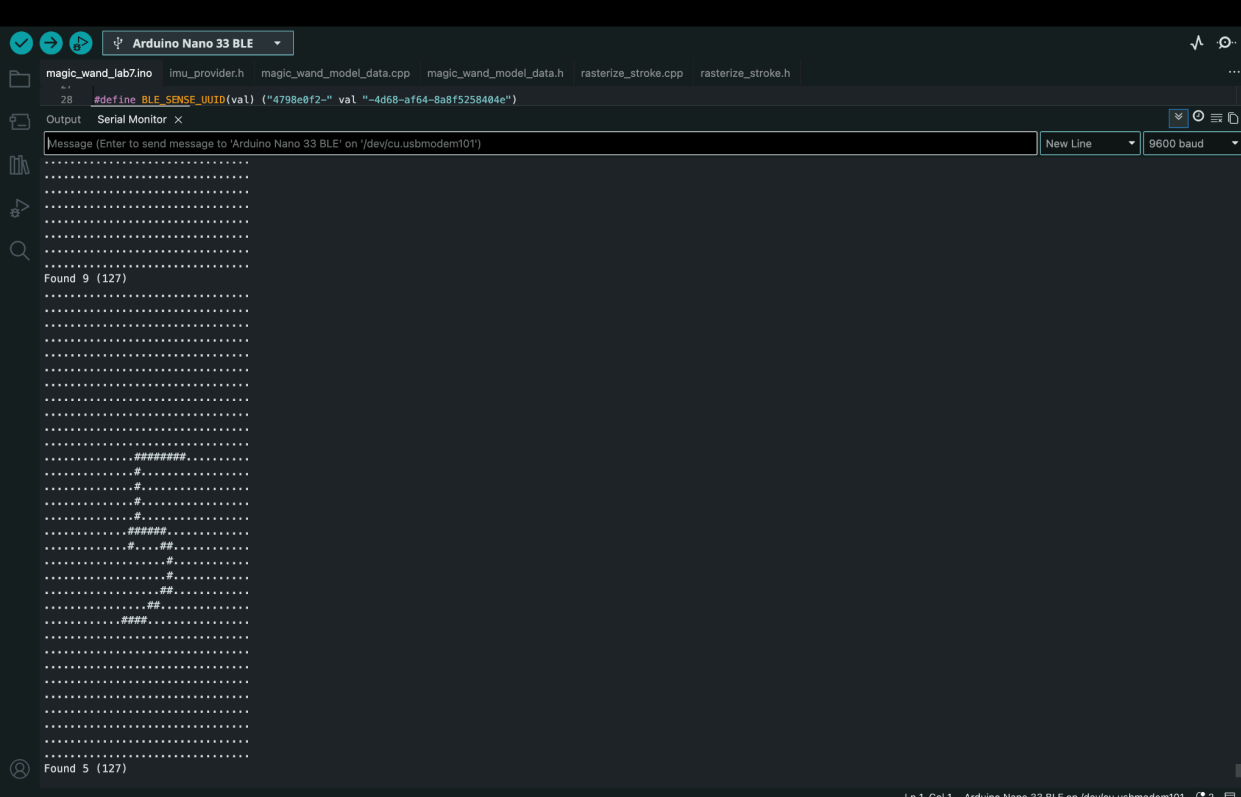


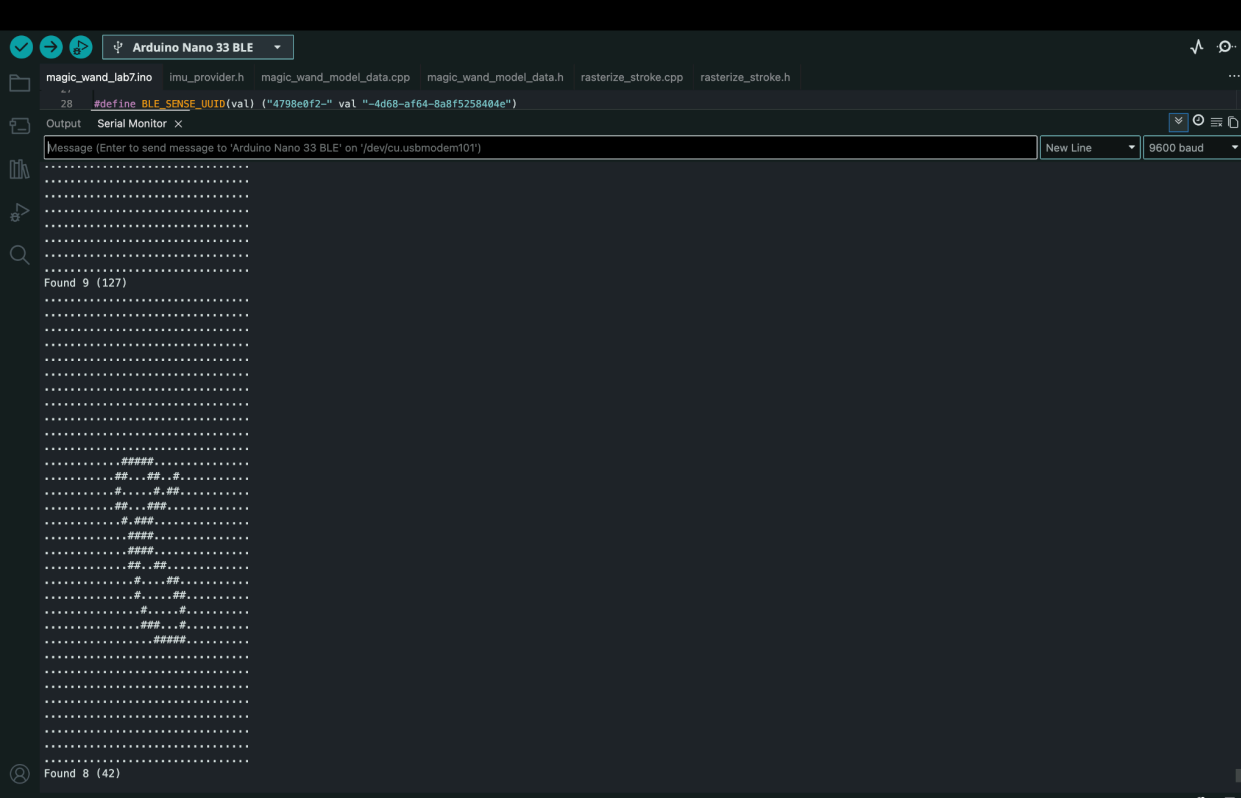
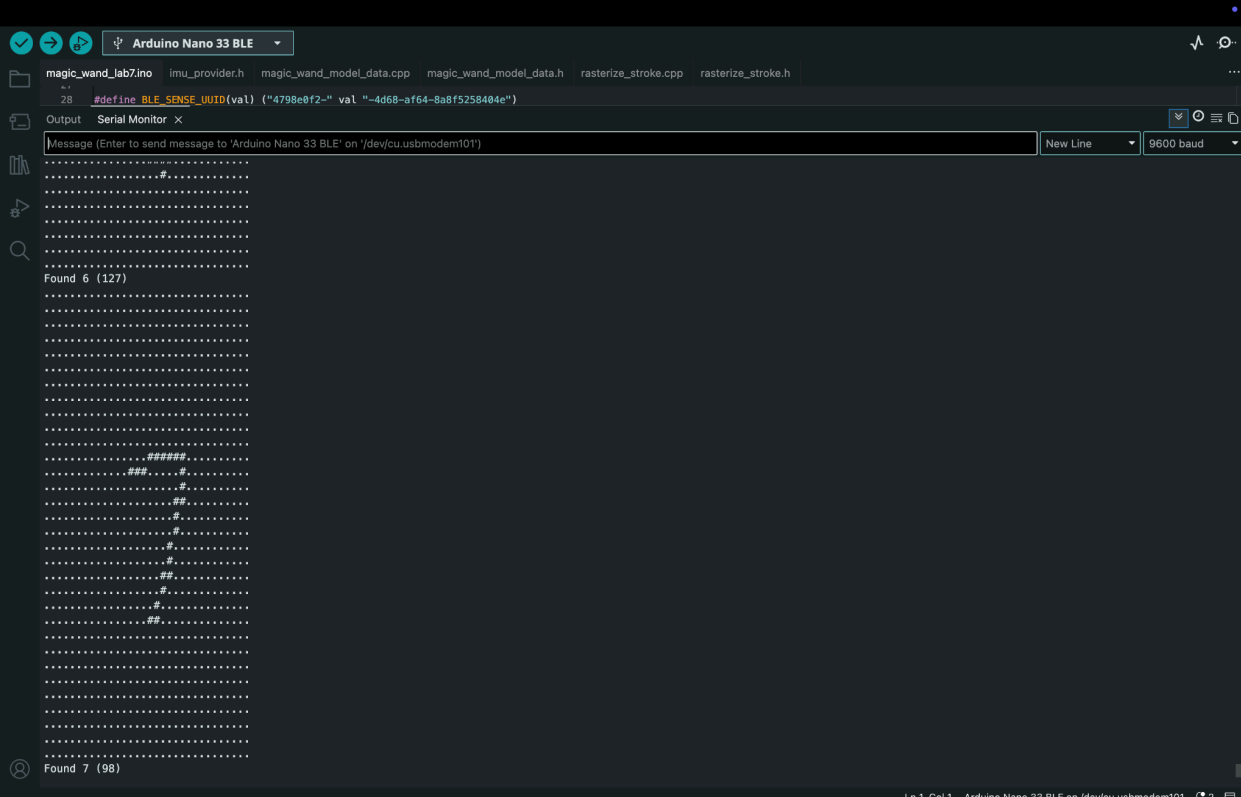


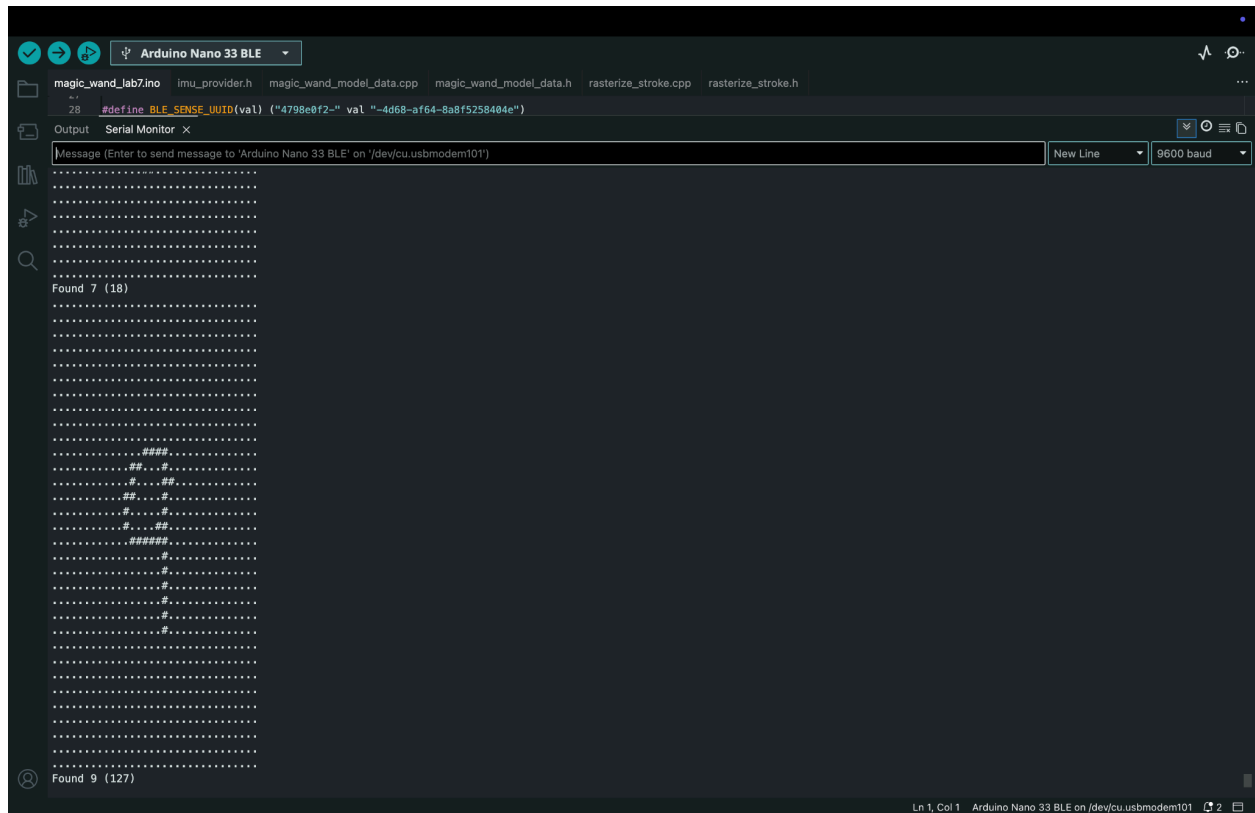
Finetuned Model











The screenshot shows the Arduino IDE interface with the 'Serial Monitor' window open. The top bar indicates the board is 'Arduino Nano 33 BLE'. The code editor shows a file named 'magic_wand_lab7.ino' with a line of code: `#define BLE_SENSE_UUID(val) ("4798e9f2-" val "-4d68-af64-8a8f5258484e")`. The Serial Monitor window shows a large amount of garbled output (noise) and a few lines of text indicating 'Found 7 (18)' and 'Found 9 (127)'. The status bar at the bottom shows 'Ln 1, Col 1' and 'Arduino Nano 33 BLE on /dev/cu.usbmodem101'.

The fine-tuned model noticeably outperformed the baseline model. With the baseline model, more digits were getting misclassified. It took a lot of attempts to get all 10 digits (0–9) recognized correctly. After fine-tuning the model with my own collected gesture data, the model was much quicker to correctly identify all 10 digits with fewer misclassifications overall.

The baseline model was trained on general gesture data that didn't necessarily match my specific drawing style, hence the difference makes sense. Fine-tuning the model with my own data let the model adapt to how I specifically draw out every digit, so it became much better at recognizing my gestures specifically. Overall, fine-tuning definitely helped; it made the deployed model more reliable and responsive to my gesture style compared to the baseline model.